Lesson 9.1: Polygon modelling, PBR texturing, lighting and rendering in Blender

What you learn

- **Polygon modelling.** Build shapes from vertices, edges and faces; keep clean quads.
- **PBR materials.** Base colour, roughness, metallic and normal maps describe how a surface reacts to light.
- **Studio lighting.** Key, fill and rim lights.
- **Rendering.** Choose Cycles for realism or Eevee for speed; denoise and set samples to balance time and quality.

Tools and resources (official sites, not checked live)

- Blender manual

ASSIGNMENT

Practice for lesson 9.1

Business problem: Model and render a mug.

Steps

1. Model the mug.
2. Apply a ceramic material.
3. Light it with three lights.
4. Render at two sample counts.

Expected output: A rendered mug and a render-time note.

SOLUTION (ANSWER KEY)

Compare the two renders to find the lowest samples that still look clean.

Check yourself

1. What does roughness control?
   Answer: How blurry the reflections are.
2. What is a key light?
   Answer: The main light.

Lesson 9.1 of the **Design and Creative** lessons. See this lesson and its official links on skillnavora.in · All chapters and lessons · Open the full lesson guide. Tools, prices and plans change; confirm on each official site.